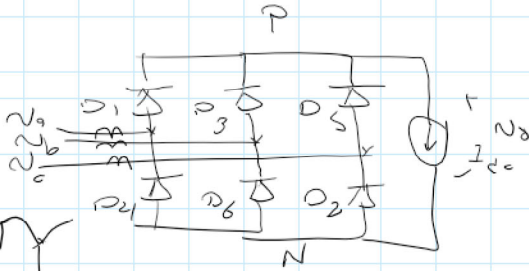


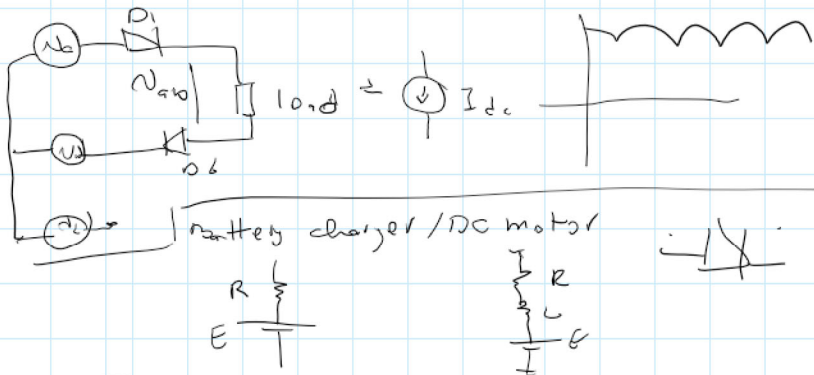
Recall

$$\bar{V}_o = \frac{3}{\pi} V_{\text{dep}}$$

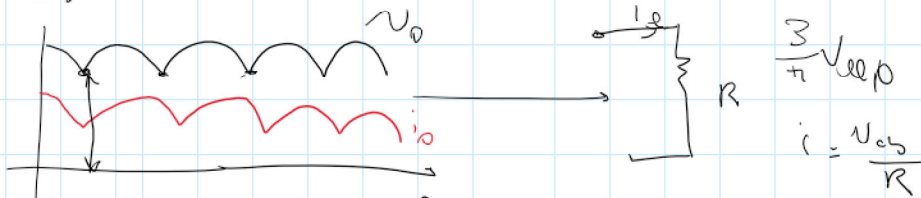


$$\text{Average } I_{dc} \text{ flow: } \bar{V}_o = \frac{3}{\pi} V_{\text{dep}} \approx V_{dc}$$

Continuous v.s. Discontinuous
equivalent circuit at each instant



resistive load



$$V_{d1} = V_{c1} \Rightarrow V_{\text{min}} = \frac{\sqrt{3}}{2} V_{\text{dep}} = 0.866 V_{\text{dep}}$$

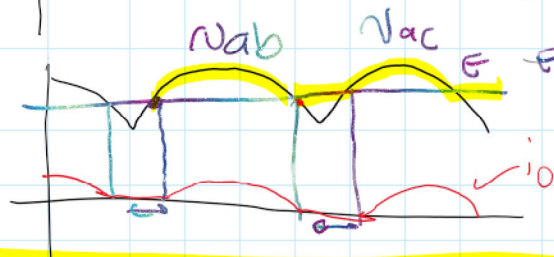
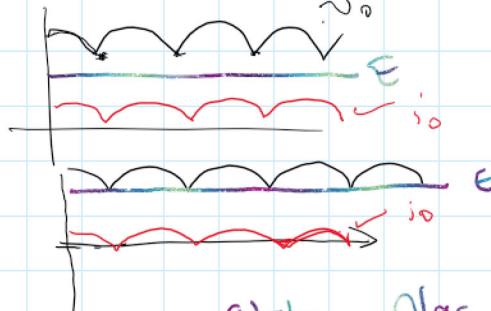
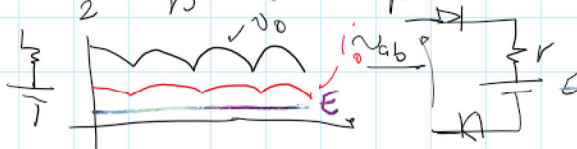
13 - Heavy charger

$$i = \frac{V_{\text{avg}} - E}{R} \quad \checkmark$$

①

③

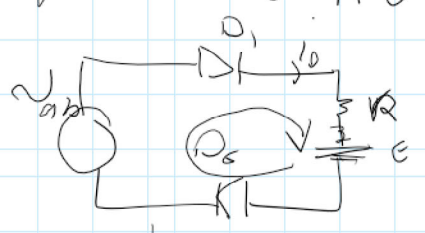
④



... ..



parameters affecting continuous operation:



$$Ri_o + L \frac{di_o}{dt} + E = V_{ab}$$

$$V_{ab} = V_{mp} \sin(\omega t + \frac{\pi}{3})$$

$N_a = V_c, N_{ac} = \phi \Rightarrow V_a = V_{mp} \sin(\omega t + \pi/3)$
 $V_{ab} = V_{mp} \sin(\omega t + \pi/3)$

$Ri_o + L \frac{di_o}{dt} + E = V_{mp} \sin(\omega t + \frac{\pi}{3})$ continuous

$i_o = i_{of}(t) + i_{on}(t)$

$i_{of}(\omega t) = \frac{V_{mp}}{Z} \sin(\omega t + \frac{\pi}{3} - \phi) - \frac{E}{R}$

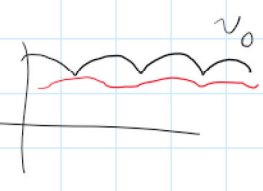
$Z = \sqrt{R^2 + \omega^2 L^2}$
 $\phi = \tan^{-1} \omega L / R$

$\frac{E}{R} = \frac{V_{mp}}{Z} \times \left(\frac{E}{V_{mp}}\right) \times \left(\frac{Z}{R}\right) = \frac{V_{mp}}{Z} \times K_E \times \frac{1}{\cos \phi}, K_E = \frac{E}{V_{mp}}$
 load EMF constant.

$i_{of}(\omega t) = \frac{V_{mp}}{Z} \left\{ \sin(\omega t + \frac{\pi}{3} - \phi) - K_E / \cos \phi \right\}$

$i_{on}(\omega t) = A e^{-t/\tau}$

$i_o(\omega t) = A e^{-t/\tau} + \frac{V_{mp}}{Z} \left\{ \sin(\omega t + \frac{\pi}{3} - \phi) - K_E / \cos \phi \right\}$



$i_o(\omega t = \phi) = i_o(\omega t = \pi/3)$

$i_o(\omega t) = \frac{V_{mp}}{Z} \left\{ \sin(\omega t + \frac{\pi}{3} - \phi) + \frac{\sin \phi}{1 - e^{-\pi/3 + \phi}} e^{-\omega t / \tau} - K_E / \cos \phi \right\}$ (1)

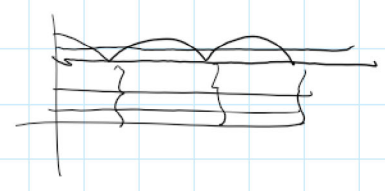
condition for continuous operation:

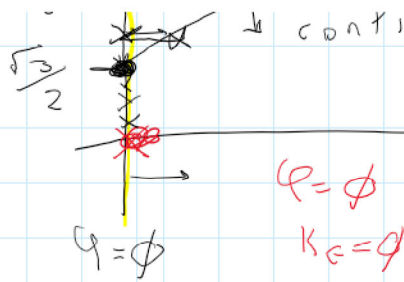
$i_o(\omega t = \phi) = i_o(\omega t = \pi/3) > 0$ *

$\sin(\frac{\pi}{3} - \phi) + \frac{\sin \phi}{1 - e^{-\pi/3 + \phi}} - \frac{K_E}{\cos \phi} > 0$ *

$K_E \leq \left[\sin(\frac{\pi}{3} - \phi) + \frac{\sin \phi}{1 - e^{-\pi/3 + \phi}} \right] \cos \phi$

$i_o(\omega t = \phi) = i_o(\omega t = \pi/3)$
 $= i_o(\omega t = 7\pi/3) \dots$

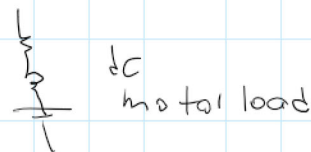




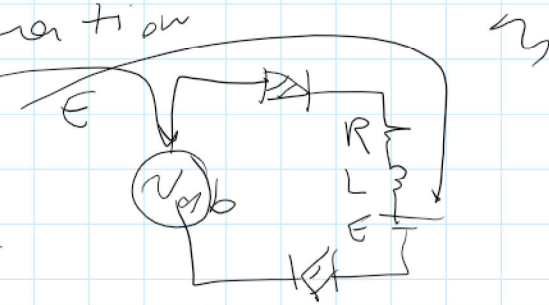
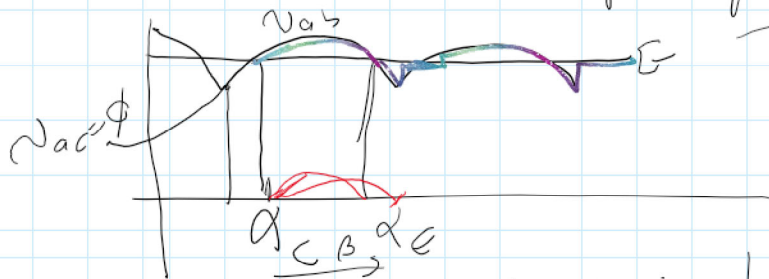
$$\begin{aligned} \phi = \phi &\Rightarrow L = \phi \\ k_e = \phi &\Rightarrow E = \phi \end{aligned} \rightarrow R$$

$$\begin{cases} k_e \neq \phi & E, R, L \\ \phi = \phi & R, L, E \end{cases} \quad \frac{1}{I} \rightarrow \text{Battery}$$

$$\begin{aligned} k_e \neq \phi & R/L/E \\ \phi \neq \phi & R/L/E \end{aligned}$$



Discontinuous mode of operation



Cross-over angle

$$R i_o + L \frac{d i_o}{dt} + E = V_{ab} = V_m \sin(\omega t + \pi/3)$$

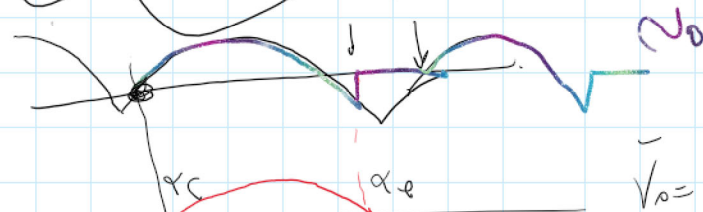
Conduction angle: $\alpha_e - \alpha_c$

$$i_o(\omega t) = \frac{V_{mp}}{Z} \left[\sin(\omega t + \pi/3 - \phi) - \frac{k_e}{\omega Z} \right] + A_0 e^{-\omega t / \tau_{sp}}$$

$$i_o(\omega t = \alpha_c) = 0$$

$$i_o(\omega t) = \frac{V_{mp}}{Z} \left\{ \sin(\omega t + \pi/3 - \phi) - \frac{k_e}{\omega Z} - \left[\sin(\alpha_c + \pi/3 - \phi) - \frac{k_e}{\omega Z} \right] e^{-\frac{\omega t - \alpha_c}{\tau_{sp}}} \right\}$$

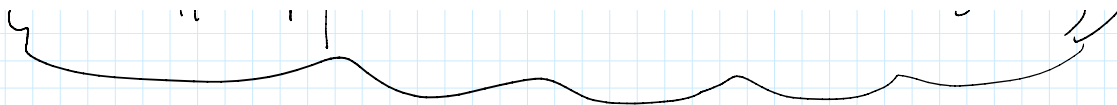
$$V_o = \frac{3}{\pi} V_{mp}$$



$$V_o = \frac{3}{\pi} V_{mp}$$

$$V_o = \frac{3}{\pi} \left\{ \int_{\alpha_c}^{\alpha_e} V_{mp} \sin(\omega t + \pi/3) d\omega t + \int_{\alpha_c}^{\alpha_c + \pi/3} E d\omega t \right\}$$

$$= \frac{3}{\pi} V_{mp} \left\{ 2 \sin(\alpha_c + \frac{\pi}{2} + \frac{\pi}{3}) \sin \frac{\beta}{2} + k_e (\frac{\pi}{3} - \beta) \right\}$$



$$\beta = \alpha_e - \alpha_c$$

$$\bar{I}_0 = \frac{\bar{V}_0 - E}{R}$$

$\alpha_c, \alpha_e, \beta$

$$V_{dp} \sin(\omega t + \pi/3) = E \Rightarrow \alpha_c = \sin^{-1}(E/V_{dp}) - \pi/3$$

$\alpha_e: i_o(\omega t) = 0 \Rightarrow \omega t = \alpha_e \Rightarrow \beta$

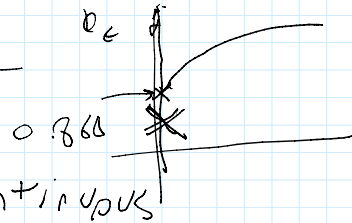
Example: 6 pulse diode rectifier
input: 3 ϕ , 230V, 50Hz

load: battery $E = 270V$, $R = 0.72\Omega$

charging current = ?

$$\bar{I}_0 = \frac{\bar{V}_0 - E}{R}$$

$$K_c = \frac{E}{V_{dp}} = \frac{270}{\sqrt{2} \times 230} = 0.83 < 0.866 \Rightarrow \text{continuous}$$



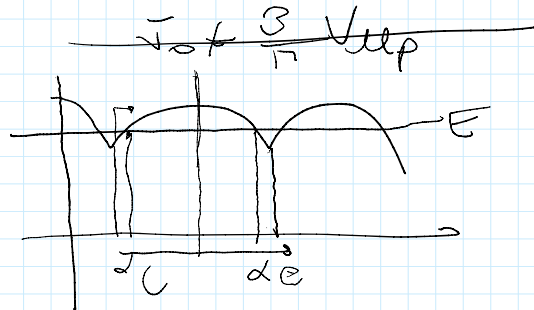
$$\bar{V}_0 = \frac{3}{\pi} V_{dp} = \frac{3}{\pi} \times \sqrt{2} \times 230 = 310.6V, \quad \bar{I}_0 = \frac{310.6 - 270}{0.72} = 55.4A$$



b) what if $E = 300V$

$$K_c = \frac{300}{\sqrt{2} \times 230} = 0.92 > 0.866 \Rightarrow \text{Discont.}$$

$$V_{dp} \sin(\omega t + \pi/3) = E \Rightarrow \alpha_c = 7.3^\circ$$



$$\alpha_e = 60 - \alpha_c = 60 - 7.3 = 52.7^\circ$$

$$\beta = \alpha_e - \alpha_c = 45.4^\circ$$

$$\bar{V}_0 = \frac{3}{\pi} V_{dp} \left\{ 2 \sin\left(\alpha_c + \frac{\pi}{2} + \frac{\pi}{3}\right) \sin \frac{\beta}{2} + K_c \left(\frac{\pi}{3} - \beta\right) \right\} = 312.73V$$

$$V_o = \frac{V}{\pi} V_{lp} \left(2 \sin \left(\alpha_c + \frac{\pi}{2} f \frac{1}{3} \right) \sin \frac{1}{2} + \cos \left(\frac{1}{3} - \beta \right) \right) = 312.73$$

$$I_o = \frac{V_o - E}{R} = \frac{312.73 - 300}{0.72} = 17.7 A$$